VG09B: Findings from the A + D variant

Date: 2026-05-14

> **Note:** This document was generated with assistance from an AI model (Claude, Anthropic) and should be independently verified.

Context

See:

- `notes/202605131400-vg09-sampler-diagnostics.md` — diagnostic investigation of VG09.
- `notes/202605131500-vg09-structural-options.md` — structural options proposed to address the q-trajectory redundancy.

VG09B implements the recommended first experiment from that note:

- **Option A (tighter q-anchor priors):** `p_slope_low_q ~ Beta(3, 22)` (mean 0.12, sd 0.06); `p_slope_hi_q ~ Beta(20, 4)` (mean 0.83, sd 0.08).
- **Option D (per-draw GP anchor):** `g_u` and `g_q` are reparameterised as `eta · (g_unit - g_unit(a_ref))` with `a_ref = 54` months, so every posterior draw of the GP passes through zero at the reference age.

All other components (study REs, subject REs on u and q, dispersion priors, likelihood) are identical to VG09.

Both models were fitted in `rep` configuration. Comparison below.

Diagnostics — every flagged parameter cleared

The structural diagnosis was correct: clearing the GP intercept ridge eliminated all marginal `r_hat` and `ess_tail` warnings, and roughly doubled bulk ESS across the previously flagged parameters.

Parameter	VG09 r_hat	VG09 ess_bulk	VG09B r_hat	VG09B ess_bulk
`slope_q`	1.020	430	**1.008**	**1244**
`p_slope_low_q`	1.013	1008	**1.007**	**1610**
`p_slope_hi_q`	1.012	431	**1.006**	**1221**
`eta_q`	1.010	483	**1.009**	**1207**
`p_slope_hi_u`	1.014	628	**1.007**	**1338**
`intercept_u`	1.012	759	**1.007**	**1225**

Parameter	VG09 r_hat	VG09 ess_bulk	VG09B r_hat	VG09B ess_bulk
`intercept_q`	1.005	799	**1.004**	**2152**

Summary line: `r_hat  1.01 and ESS  400 across reported parameters.`

Three-way comparison of q-trajectories (VG07, VG09, VG09B)

VG07 is the relevant baseline because it has no subject REs on q and therefore is not subject to the GP RE intercept redundancy that motivates VG09B.

Overlay plot and merged CSV are written by `scripts/compare_models.py` to:

- `output/comparisons/vg07_vg09_vg09b_q_by_age.{png,svg}`
- `output/comparisons/vg07_vg09_vg09b_q_by_age.csv`

Age (mo)	VG07 q_median	VG09 q_median	VG09B q_median
12	0.041	0.011	0.007
18	0.085	0.033	0.021
24	0.148	0.073	0.048
30	0.211	0.133	0.089
36	0.300	0.260	0.182
42	0.443	0.474	0.363
48	0.599	0.657	0.534
54	0.711	0.769	0.651
60	0.782	0.846	0.757
66	0.833	0.890	0.829
72	0.871	0.898	0.844
78	0.904	0.882	0.820
84	0.934	0.870	0.804
90	0.958	0.882	0.825

Patterns

1. **VG07 is monotone-increasing across the whole age range** (0.04 → 0.96). This is the expected shape of `p_S / p_U` for DS in published CDI data.
2. **VG09 and VG09B both peak then dip** at the upper tail: VG09 peaks ~0.90 at 72 mo then falls to ~0.88 at 90 mo; VG09B peaks ~0.84 at 72 mo and falls to ~0.82 at 90 mo. The non-monotone shape is a consequence of adding subject REs on q — extrapolation into ages with sparse data is dragged toward the prior by the subject-RE marginalisation, and the GP can bend downward.

3. **At mid-ages (36–66 mo) VG07 sits between VG09 (highest) and VG09B (lowest).** VG09B's level is roughly 10 percentage points below VG07 and 10–13 percentage points below VG09. At 48 mo, VG07/VG09/VG09B are 0.60 / 0.66 / 0.53.
4. **At very young ages (12–30 mo) VG07 is much higher than both VG09 and VG09B.** This is a side-effect of the subject REs: with a wider subject-level distribution in logit-q space, the population-level latent `q` at the lower edge of the data range shrinks toward zero (Jensen-effect on the inverse-logit transform).

What changed between VG09 and VG09B

The reparameterisation moved `intercept_q` from −1.17 to −0.23 (an upward shift of about 0.94 logit units) and `p_slope_hi_q` from 0.80 to 0.93. `tau_subj_q` is essentially unchanged (~1.19) and `eta_q` is essentially unchanged (~0.28). The shift is therefore entirely between the GP and the linear trend, not between the GP and the REs.

Interpretation: in VG09 the unanchored `g_q` carried roughly 1.5 logit units of constant level at the reference age. The data could not distinguish that constant from `intercept_q + slope_q · a_z(a_ref)`, which is exactly the ridge identified in the structural-options note. Once the GP is forced to pass through zero at 54 months, the linear trend uniquely defines the level there and the GP is restricted to describing genuine deviations from linearity.

The downward shift in q at mid-ages is a real consequence of removing that hidden level: without the GP contributing a positive constant on every draw, the maximum-a-posteriori q at any age is lower than it was under VG09.

Reading the three-way comparison

There are two distinct comparisons hiding inside this.

VG07 → VG09: adds subject REs on q. The change is most visible at the young end (the shrinkage to zero at 12–24 mo) and at the upper tail (the plateau / dip after 72 mo). The mid-age levels are broadly similar.

VG09 → VG09B: the parameterisation fix. The change is most visible at mid-ages (the downward shift of ~5–10pp) because that is where the GP-level redundancy was being absorbed. The young-age and upper-tail shapes are similar to VG09.

VG09B is *not* "VG09 made to look like VG07". It is "VG09 with the latent GP-level identifiability resolved". The remaining difference between VG09B and VG07 is the genuine effect of adding subject REs on q.

Implications for the model family

- **VG09B's diagnostics are clean and the parameterisation is structurally honest.** It is a defensible candidate to replace VG09.
- **If we replace VG09 with VG09B, we should also apply the GP anchor to `g_u`/`g_q` in VG05–VG08** (and to `g` in the univariate VG01–VG04) for consistency. Those models have the same GP intercept ridge in principle; we just haven't seen it bite the diagnostics yet because they have fewer overlapping global components on each outcome.
- **The mid-age q estimates change substantively** (~5–10pp lower in VG09B than VG09). This is not just a parameterisation cleanup — the reported trajectory will move. The shift is in the direction supported by the structural argument: VG09's higher mid-age q values were partly an artefact of the GP-level redundancy, not a fact about the data.
- **The non-monotone upper tail is shared by VG09 and VG09B** and is the most important interpretive question that remains. It reflects how subject REs on q interact with the GP at ages where the data are sparse and the population mean is dominated by prior pull. Worth a separate investigation: either tighten `tau_subj_q`, narrow the GP lengthscale prior, or restrict the query range to ages with substantive data.

Suggested next steps

1. Decide whether to promote VG09B → VG09. The case is strong: diagnostics resolved, parameterisation is the standard fix, mid-age shift is defensible. Cost: the technical report needs revising and the published mid-age q values will move.
2. If promoting, apply the GP anchor to `common_bivariate.py` (VG05–VG08) and `common.py` (VG01–VG04) symmetrically.
3. Separately investigate the non-monotone upper tail in q. This is independent of A+D and would apply to VG07, VG09, and VG09B equally.

See `output/models/VG09B-age-understood-spoken-ds-re-subj-uq-anchored/` for the full VG09B output and report.